Amine Maazizi

PhD Researcher in 3D Computer Vision and Representation Learning | Mines Paris – PSL
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Website | GitHub | LinkedIn

Research Interests

My research interests lie in 3D computer vision, representation learning, and spatial intelligence. I am particularly interested in developing generalizable 3D representations for open-vocabulary scene understanding in unstructured environments and exploring their applications to embodied perception.

Education

Mines Paris – PSL, Université PSL **Starting Nov 2026**
PhD in 3D Computer Vision and Robotics

ENS Paris-Saclay **2025 – 2026**
Master in Mathematics, Vision and Learning (MVA)
Relevant coursework: Geometric Deep Learning, Geometry Processing, Geometric Data Analysis, Representation Learning, Graphs in Machine Learning, Multimodal Explainable AI.

ENSTA Paris, Institut Polytechnique de Paris **2023 – 2026**
Engineering Degree in Computer Science and Artificial Intelligence GPA: **4.28/4.3**, Top 5%
Relevant coursework: Machine Learning, Statistical Learning, Deep Learning, Computer Vision, Language Models, Reinforcement Learning.

CPGE Moulay Youssef, Rabat, Morocco **2021 – 2023**
Preparatory Classes (MPSI / MP)* Top 3% nationally

Research Experience

Mines Paris – PSL, Centre for Robotics, Paris, France **Starting Nov 2026**
Incoming Doctoral Researcher | Collaboration with ENSTA, U2IS (Palaiseau)
Supervisors: Prof. Jean-Emmanuel Deschaud (Mines Paris – PSL); Prof. François Goulette and Prof. Philippe Xu (ENSTA)
Research on transferring knowledge from vision and vision-language foundation models to LiDAR point clouds for open-vocabulary semantic segmentation in unstructured terrain, with extensions to robot-specific affordance learning.

EPFL, Artificial Intelligence in Molecular Medicine Lab, Lausanne, Switzerland **Apr 2026 – Sep 2026**
Research Intern
Supervisors: Dr. Lukas Klein and Prof. Charlotte Bunne
Developed VirTues-sim, a latent-space perturbation model built on a spatial-proteomics foundation model for treatment-response simulation.

Institut Pasteur, Paris, France **Jun 2025 – Aug 2025**
Research Intern
Supervisor: Prof. Giacomo Nardi
Developed a non-rigid 3D mesh-matching method incorporating mechanical constraints to support morphodynamic analysis and force estimation during endothelial-to-hematopoietic transition.

NAIST, Cybernetics and Reality Engineering Lab, Remote **Oct 2024 – Dec 2024**
Research Collaborator
Supervisor: Prof. Hideaki Uchiyama
Conducted a literature review of multimodal 3D generation methods.

Selected Research Presentation

α-CBM: Label-Free Concept Bottleneck Models for Audio
Amine Maazizi, Adam Gassem.
Oral presentation, Journées de la Science des Données et de l'Ingénierie (JDSE), 2026.

Open-Source Software

PyMIWAE **GitHub / PyPI**

PyTorch package implementing MIWAE, not-MIWAE, supMIWAE, and our supervised MNAR extension in a unified framework.

Awards and Honors

Finalist, Best Research Project Award, ENSTA Paris **2025**

Moroccan Excellence Scholarship, Ministry of Higher Education **2023**

Technical Skills

Programming: Python, C, C++
Machine Learning: PyTorch, JAX, TensorFlow, PyTorch Geometric, Scikit-learn
Scientific Computing: NumPy, SciPy
Infrastructure: Linux, Git, HPC (SLURM)

Languages

Arabic (Native) | French (Bilingual) | English (Fluent) | Spanish (Elementary)